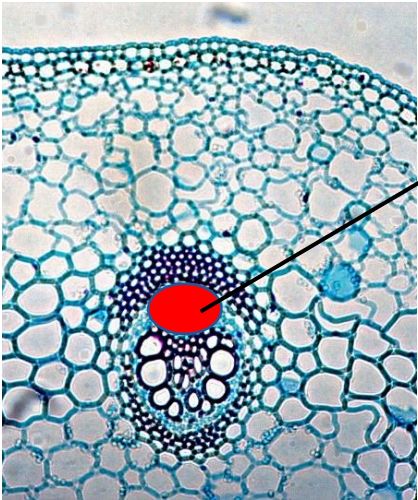


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)	(i)		Lignin (1) To prevent the collapse of the xylem/ provide (mechanical) support (1) ignore strength/ waterproofing/ rigidity	2			2		1
		(ii)		Any value between 238-254 μm = 2 marks If incorrect award 1 mark for Any value between 0.238-0.254mm measurement/130		2		2	2	
		(iii)		Any value between 0.85-0.91 = 3 marks If incorrect award 2 marks Any value between 238-254/280 Any value between 0.238-0.254/ 0.280 If incorrect award 1 mark for sight of Any value between 238-254/0.28 Any value between 878-879 ECF from (ii) Award 1 mark for: answer from aii/0.28 Award 2 marks for: answer from aii/280 or (answer from aii/1000)/0.28 Award 3 marks for: correct calculation from above		3		3	3	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			A. Xylem contents under lower pressure (than air) (1) B. contents pulled (upwards through the xylem) due to {transpiration stream / cohesion-tension / adhesion/ evaporation of water vapour from the leaves} (1) C. Phloem contents under higher pressure (than air) (1) D. contents {pushed/forced} through phloem by {mass flow/ active transport/ hydrostatic pressure} (1)		4	0	4		4
	(c)	(i)		Amino acid/named amino acid / (named) hormone (1)	1			1		
		(ii)		Any four (x1) from: A. There are (high levels of) radioactivity in young leaves (1) B. Young leaves are sinks (1) C. so {using/importing} sucrose/ sucrose transported to young leaves (1) D. (as an energy source) for {growth/cell division} (1) E. Lack of radioactivity in older leaves as {they are sources/ produce their own sucrose} (1)		2	2	4		4
		(iii)		Bidirectional flow/ both up and down (the stem)/ both directions		1		1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		Phloem labelled in this region 		1		1		1
				Question 5 total	3	13	2	18	5	7